Input Saturated Visual Servoing for Unmanned Aerial Vehicles

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***Abstract*—This paper presents an input saturated visual servoing controller for a quadrotor unmanned aerial vehicle (UAV). The controller regulates relative pose between the vehicle and a planar horizontal visual target. In order to simplify the control design, the method uses a virtual camera to define a set of image moment features. The image feature kinematics is independent of roll and pitch motion. Since the image feature error is stabilized in the virtual camera, the visual target potentially leaves the real camera's field of view (FoV). To keep the visual target in the camera FoV, an input saturated law is proposed to sufficiently bound the roll and pitch of the vehicle. Furthermore, an adaptive input saturated control law is proposed to ensure positive thrust. Experimental and simulation results are provided to demonstrate controller performance.**

***Index Terms*—Bounded control, image-based visual servoing (IBVS), image moments, virtual camera.**

I. Introduction

VISUAL servoing of unmanned aerial vehicles (UAVs) involves using feedback from computer vision to perform motion control. Visual servoing techniques allow a UAV to fly in GPS-denied environments [1] or can be applied to control relative position to a visual target as in autonomous landing [2]. In this paper, we consider visual servoing of a quadrotor UAV with a single downward facing camera and the proposed visual servoing law regulates the quadrotor's relative pose to a stationary planar horizontal target. Visual servoing laws usually are categorized into two approaches: 1) position-based visual servoing (PBVS); and 2) image-based visual servoing (IBVS) [3]. In this paper, we focus on IBVS methods where the error signal to be stabilized is directly defined in the image plane. Unlike PBVS, IBVS avoids the need for pose reconstruction which requires accurate camera calibration and a prior three-dimensional (3-D) model of the target. In general, IBVS is simple to implement and is relatively insensitive to camera calibration [4].

Previous work on IBVS for UAVs can be classified into four categories: 1) the spherical image moment-based design [5], [6]; 2) homography-based methods [7], [8]; 3) the virtual spring approach [9], [10]; and 4) the virtual camera approach [11]–[14]. A summary of these approaches can be found in our previous work [15]. In this paper, we follow [11]–[14] and employ a virtual camera that has the same position and yaw as the real camera but with zero roll and pitch. The image features for visual servoing are defined in the virtual camera image plane. Hence, a transformation is required to convert the image features in the real camera image plane into the image feature in the virtual camera plane. The expression for this transformation can be derived from geometry as mentioned in [12] and [13], or by solving a system of first-order homogeneous partial differential equations [15]. The transformation depends on an attitude measurement that is commonly available from an onboard attitude and heading reference system (AHRS). Since the virtual camera has zero roll and pitch motion, the image feature kinematics is independent of roll and pitch rates. This independence is crucial for simplifying the image feature kinematics and facilitating the control design.

Although the image feature error defined in the virtual camera image can be stabilized, there is no guarantee that the visual target remains in the real camera's field of view (FoV). Hence, the control gains in our previous work [13], [15], [16] were chosen small to avoid aggressive motion that makes the visual target leave the camera FoV. The problem of keeping the visual target in the camera FoV has been considered in the literature. In [17], an output constraint method from [18] is used for landing a UAV with parallel line image features. An alternative approach is to constrain the vehicle's attitude (e.g., [19] and [20]). Although this approach has no rigorous proof that the visual targets remain in the camera FoV, it is simpler to implement relative to [17]. In this paper, we propose a new control law based on an input constraint method.

The proposed approach uses an inner–outer loop control structure. The inner loop tracks an attitude reference from the outer IBVS loop that regulates the image feature error in the virtual camera image. We assume the inner loop has been appropriately designed and focus on the outer loop IBVS design. The main contribution of this paper is a new nested saturation control for the outer loop to ensure the roll, pitch, and thrust inputs lie within specified bounds. Bounds on roll and pitch keep the visual target in the camera FoV, and positive thrust is a constraint for typical quadrotors. Compared with the commonly used nested saturation approach mentioned in [21] and its variations such as [22] and [23], the proposed method benefits from being able to use less conservative (i.e., lower) input bounds

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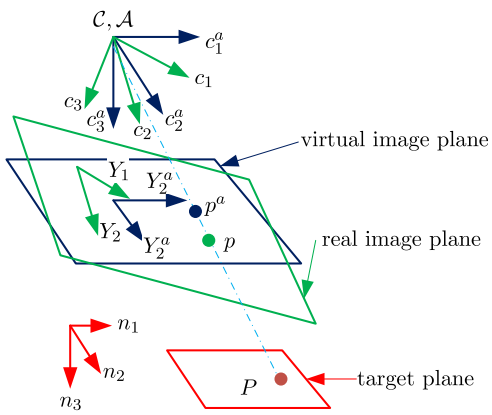

Fig. 1. Frame definitions and the pinhole camera model.

when input disturbances are present. As discussed in [24], the reason for this improved performance is due to not requiring the state transformation used in [21]. In this paper, the propeller thrust is modeled by the product of a thrust gain and the square of the physical input: pulse-width modulation (PWM) voltage to the electronic speed controller (ESC). As mentioned in our previous work [15], the thrust gain gradually decreases as the battery voltage drops, and this noticeably affects the UAV motion. Another contribution of this paper is that the nested saturation IBVS can compensate for variation in thrust gain. Furthermore, the proposed control law is also adaptive to uncertainty in mass and desired depth. As is well known, the proposed IBVS law provides high gain when the system state approaches its equilibrium. This high-gain control gain helps to reduce the steady-state error resulting from bias attitude measurement that is considered in [25]. In order to validate the performance of the proposed control, we implement it in simulation and experiment. This should be compared to the related work [19], [20] where only simulation results are provided.

This paper is organized as follows. Section II describes the camera model, image features, kinematics of the image features, and UAV dynamics. Section III gives the details of the input saturation controller design and the proof of stability for the closed-loop. Section IV shows both simulation and experimental results using the proposed control. Finally, concluding remarks are provided in Section V.

II. Dynamic Visual Servoing Modeling

A. Reference Frames

As shown in Fig. 1, we first introduce three coordinate frames used for modeling. A navigation frame $\mathcal{N}$ is assumed to be inertial and fixed to a point on the Earth with a basis $\{n_1, n_2, n_3\}$ oriented north, east, and down, respectively. A downward looking camera is rigidly attached to the quadrotor and the camera's optical center is assumed to coincide with the center of mass (CoM) of the vehicle. Hence, a camera frame $\mathcal{C}$ is introduced with its origin fixed at the camera's optical center and basis $\{c_1, c_2, c_3\}$ oriented forward, right, down, respectively. The vector $d_{nc}^n = [d_1, d_2, d_3]^T$ denotes the displacement of the origin of $\mathcal{C}$ in $\mathcal{N}$. The orientation of $\mathcal{C}$ relative to $\mathcal{N}$ is represented by a rotation matrix R, which is parametrized by the *ZYX* Euler angles $\eta = [\phi, \theta, \psi]^T$ as $R = R_{n_3}(\psi) R_{n_2}(\theta) R_{n_1}(\phi)$, where R_{n_k} denotes an elementary rotation about the n_k axis. We note that the Euler angle representation is singular at $\theta = (2k+1)\pi/2$, $k \in \mathbb{Z}$. However, singularities are normally encountered only in acrobatic flight that is not typical in visual servoing. Lastly, a virtual camera with frame $\mathcal{A}$ is introduced. The frame $\mathcal{A}$ has the same origin as $\mathcal{C}$ and a basis $\{c_1^a, c_2^a, c_3^a\}$, where c_3^a is aligned with the virtual camera's optical axis. The virtual camera has zero roll and pitch, which means that c_3^a is parallel to n_3.

B. Image Feature and Kinematics

In this paper, the pinhole camera model is used, as shown in Fig. 1. In the image plane, a 2-D image frame $\mathcal{Y}$ is introduced with basis $\{Y_1, Y_2\}$ parallel to c_1 and c_2, respectively. A 3-D point with coordinates $P = [X_1, X_2, X_3]^T$ in $\mathcal{C}$ is projected onto a point in the image plane with coordinates $p = [y_1, y_2]^T = \lambda[X_1/X_3, X_2/X_3]^T$, where λ is the focal length of the camera. The 3-D point expressed in $\mathcal{A}$ is given by $P^a = [X_1^a, X_2^a, X_3^a]^T$ and its image in the virtual image plane is $p^a = [y_1^a, y_2^a]^T = \lambda[X_1^a/X_3^a, X_2^a/X_3^a]^T$. The transformation from p to p^a can be found in [15].

Assumption 1: The target is planar and parallel to the n_1-n_2 plane. The target consists of $N > 1$ points.

As mentioned in [15], the features for visual servoing adopted here are

$$s_{lp} = [y_{1g}, y_{2g}]^T s_{hp} \tag{1a}$$

$$s_{hp} = \sqrt{a^*/a} \tag{1b}$$

$$s_{\psi p} = \arctan\left(2\mu_{11}/(\mu_{20} - \mu_{02})\right)/2 \tag{1c}$$

where $y_{1g} = \frac{1}{N}\sum_{k=1}^{N} y_{1k}^a$, y_{ij}^a is the ith component of the jth point, $y_{2g} = \frac{1}{N}\sum_{k=1}^{N} y_{2k}^a$, $a = \mu_{20} + \mu_{02}$, $\mu_{ij} = \sum_{k=1}^{N}\left(y_{1k}^a - y_{1g}^a\right)^i\left(y_{2k}^a - y_{2g}^a\right)^j$, and a^* is the value of a when the vehicle is in its desired pose. The features s_{lp}, s_{hp}, and $s_{\psi p}$ are used to control the vehicle's lateral displacement, height, and yaw motion, respectively. The kinematics using features (1) is

$$\dot{s}_{lp} = -\lambda v_l^a/X_3^{a*} - S s_{lp}\dot{\psi} \tag{2a}$$

$$\dot{s}_{hp} = -v_3^a/X_3^{a*} \tag{2b}$$

$$\dot{s}_{\psi p} = -\dot{\psi} \tag{2c}$$

where X_3^{a*} is the desired value of X_3^a, $S = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, $v_l^a = [v_1^a, v_2^a]^T$, and $v^a = [v_1^a, v_2^a, v_3^a]^T$ is camera's linear velocity expressed in $\mathcal{A}$.

C. Quadrotor Dynamics

We write the quadrotor's translational dynamics in $\mathcal{A}$ and its rotational dynamics in $\mathcal{C}$ as

$$\dot{v}^a = -\left[\dot{\psi}E_3\right]_\times v^a + g n_3 + F^a/m \tag{3a}$$

$$\dot{R} = R\left[\omega^c\right]_\times \tag{3b}$$

$$\dot{\omega}^c = -J^{-1}\left[\omega^c\right]_\times J\omega^c + J^{-1}\tau^c \tag{3c}$$

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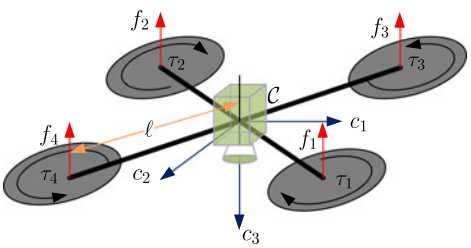

Fig. 2. Input forces f_i and torques τ_i acting on the quadrotor.

where $E_3 = [0, 0, 1]^T$, g is the gravity constant, m is the mass of the vehicle, and F^a is the applied force on the vehicle expressed in $\mathcal{A}$

$$[x]_\times = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$$

ω^c is the angular velocity of the camera, J is the inertia of the vehicle, and τ^c is the applied torque on the vehicle expressed in $\mathcal{C}$. As shown in Fig. 2, the quadrotor has four fixed-pitch propellers in a cross configuration with adjacent propellers rotating in opposite directions. Each propeller creates thrust f_i and countertorque τ_i. We assume the four motors are identical, and approximate $f_i = K_T \tilde{W}_i^2$, $\tau_i = K_\tau \tilde{W}_i^2$, where K_T, K_τ are aerodynamic constants, $\tilde{W}_i = W_i - W_{\min}$, W_i is the PWM signal sent to the ESC driving ith motor, and $W_{\min}$ is the minimum pulse width to rotate the motor. The applied force F^a and torque τ^c are given by

$$F^a = -T_M R_{\theta\phi} c_3 \tag{4a}$$

$$\tau^c = \begin{bmatrix} \ell\left(f_2 + f_3 - f_1 - f_4\right)/\sqrt{2} \\ \ell\left(f_1 + f_3 - f_2 - f_4\right)/\sqrt{2} \\ \sum_{i=1}^4 (-1)^i \tau_i \end{bmatrix} \tag{4b}$$

where $T_M = \sum_{i=1}^4 f_i$ is the thrust created by all propellers, $R_{\theta\phi} = R_{n_2}(\theta) R_{n_1}(\phi)$, and ℓ is the distance from the motor to the CoM of the vehicle. Here, we choose commonly used simple models for F^a and τ^c [5], [15]. In visual servoing of UAVs, the speed of the vehicle is slow and aggressive motion is avoided. The vehicle usually flies near a hover configuration. As pointed in [26] and [27], in hover small body forces and drag forces are negligible, and gyroscopic effects due to propeller inertia cancel. Hence, small body forces, drag forces, and gyroscopic forces are ignored in the model. The model chosen has the benefit of leading to a simple control law as mentioned in [15].

III. Controller Design

A. Saturation Functions

To present the control design this section presents three saturation functions. The first saturation function $\sigma_i : \mathbb{R} \to \mathbb{R}$ has the following properties.

1) $x\sigma_i(x) > 0, x \in \mathbb{R} \setminus \{0\}$.
2) $0 \le \frac{d\sigma_i}{dx}(x) \le k_i, x \in \mathbb{R}$.
3) $\sigma_i(x) = k_i x, |x| \le l_i$.
4) $|\sigma_i(x)| \le M_i, |x| \ge l_i$.

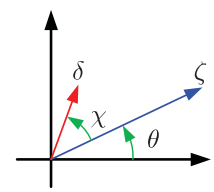

Fig. 3. Vectors δ and ζ.

The controller design given below uses three saturation functions $\sigma_i, i = 1, 2, 3$ and the conditions for closed-loop asymptotic stability will depend on their parameters $l_i, k_i, M_i, i = 1, 2, 3$. The second saturation function $\Sigma_i : \mathbb{R}^2 \to \mathbb{R}^2$ is defined as

$$\Sigma_i(\zeta) = [\sigma_i(\zeta_1), \sigma_i(\zeta_2)]^T$$

where $\zeta \in \mathbb{R}^2$ and $\zeta = [\zeta_1, \zeta_2]^T$.

Lemma III.1: Given $\delta, \zeta \in \mathbb{R}^2$, if $\|\delta\| < l_i/2$ and $\|\zeta\| > l_i/2$, where $\|\cdot\|$ is the Euclidean norm, then $\zeta^T \Sigma_i(\zeta + \delta) > 0$

Proof: The vector ζ is parametrized by θ and α_ζ as $\zeta = \alpha_\zeta l_i [c_\theta, s_\theta]^T/2$ where $|\alpha_\zeta| > 1$, $c_x = \cos x$, and $s_x = \sin x$. As shown in Fig. 3, the angle between δ and ζ is χ, and the vector δ can be written as

$$\delta = \frac{\alpha_\delta l_i}{2}\left(c_\chi \begin{bmatrix} c_\theta \\ s_\theta \end{bmatrix} + s_\chi \begin{bmatrix} s_\theta \\ -c_\theta \end{bmatrix}\right)$$

where $|\alpha_\delta| < 1$. Then, we have

$$\zeta^T \Sigma_i(\zeta + \delta) = \alpha_\zeta l_i \varphi(\theta, \alpha_\zeta, \chi, \alpha_\delta)/2$$

where $\varphi(\theta, \alpha_\zeta, \chi, \alpha_\delta) = \begin{bmatrix} c_\theta & s_\theta \end{bmatrix} \Sigma_i(\zeta + \delta)$. Proving $\zeta^T \Sigma_i(\zeta + \delta) > 0$ is equivalent to proving $\varphi(\theta, \alpha_\zeta, \chi, \alpha_\delta) > 0$. For the case $0 \le \theta < \pi/2$, from the nondecreasing Property 2) of σ_i we have

$$\varphi(\theta, \alpha_\zeta, \chi, \alpha_\delta) \ge \begin{bmatrix} c_\theta & s_\theta \end{bmatrix} \Sigma_i\left(\frac{l_i}{2}\begin{bmatrix} c_\theta + \alpha_\delta\left(c_\chi c_\theta + s_\chi s_\theta\right) \\ s_\theta + \alpha_\delta\left(c_\chi s_\theta - s_\chi c_\theta\right) \end{bmatrix}\right).$$

Because $|\alpha_\delta| < 1$, we have $c_\theta + \alpha_\delta\left(c_\chi c_\theta + s_\chi s_\theta\right) \le 2$ and $s_\theta + \alpha_\delta\left(c_\chi s_\theta - s_\chi c_\theta\right) \le 2$. We conclude that every element of the function Σ_i lies in the linear region and

$$\varphi(\theta, \alpha_\zeta, \chi, \alpha_\delta) \ge l_i\left(1 + \alpha_\delta c_\chi\right)/2.$$

Since $|\alpha_\delta| < 1$, we have $\varphi(\theta, \alpha_\zeta, \chi, \alpha_\delta) > 0$. For the case $\pi/2 \le \theta < 2\pi$, a similar procedure to that presented above can be used to show $\varphi(\theta, \alpha_\zeta, \chi, \alpha_\delta) > 0$. This proves $\zeta^T \Sigma_i(\zeta + \delta) > 0$. ■

The third saturation function introduced is $h : \mathbb{R}^2 \to \mathbb{R}^2$ defined as

$$h(\zeta) = \zeta/(1 + \zeta^T \zeta), \quad \zeta \in \mathbb{R}^2. \tag{5}$$

It can be easily shown that $\|h(\zeta)\| < 1, \zeta \in \mathbb{R}^2$.

B. Inner–Outer Loop Control Structure

Similar to the work [15] and [16], we adopt an inner–outer loop structure in this paper, as shown in Fig. 4. The inner loop tracks an attitude reference denoted by R^*, and the outer loop provides this reference to the inner loop in order to regulate

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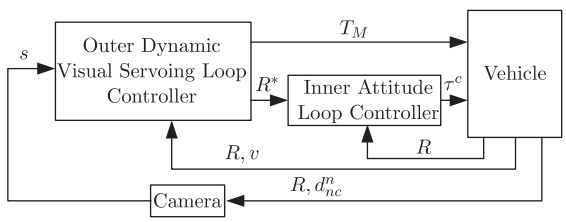

Fig. 4. Inner–outer control loop structure.

the image feature error. A main reason for using this structure is that it provides a reliable and simple solution for motion control. Each loop is effectively fully actuated and this simplifies the control design. The visual servoing problem naturally fits this structure since the AHRS measurements are relatively fast compared to the image feedback. The relatively low sampling rate of the image due to its acquisition and processing limits the bandwidth of the outer loop. The stability of the closed-loop system requires the inner loop to have a relatively high bandwidth, which is usually achieved by a high-gain controller. A stability analysis of this structure can be found in [28] and [15]. In this paper, we only consider the controller design for the outer loop and assume the inner loop has sufficient bandwidth to ensure the stability of the closed-loop system.

C. Outer Loop Controller Design

To keep the target in the camera FoV using the virtual camera approach, roll and pitch are kept small. Therefore, using the small angle approximation, the expression of F^a in (4a) becomes

$$F^a = K_T u[-s_\theta c_\phi, s_\phi, -c_\theta c_\phi]^T \approx K_T u[-\theta, \phi, -1] \tag{6}$$

where $u = \sum_{i=1}^4 \tilde{W}_i^2$. The control design is divided into two components: the height control, and the control for lateral and yaw motion.

1) Height Control: Using (2b), (3a), and (6), we have the height subsystem

$$\dot{s}_{hp} = -b_1 v_3^a \tag{7a}$$

$$\dot{v}_3^a = b_2(d_g - u) \tag{7b}$$

where $b_1 = 1/X_3^{a*}, b_2 = K_T/m$, and $d_g = mg/K_T$. We treat b_1, b_2, and d_g as unknown constants, and assume d_g lies in the compact set

$$\Omega_{dg} := \{d_g : |d_g - d_{g0}| \le d_{gm}\}$$

where d_{g0} is an initial estimate of d_g, and every element in Ω_{dg} is greater than zero. Defining

$$\delta_{h1} = s_{hp} - 1 \tag{8a}$$

$$\delta_{h2} = v_3^a - \sigma_1(\delta_{h1}) \tag{8b}$$

and using (7) we have

$$\dot{\delta}_{h1} = -b_1(\sigma_1(\delta_{h1}) + \delta_{h2}) \tag{9a}$$

$$\dot{\delta}_{h2} = b_2(d_g - u) + b_1 \frac{d\sigma_1}{d\delta_{h1}}(\sigma_1(\delta_{h1}) + \delta_{h2}). \tag{9b}$$

For the subsystem (7), the control law is chosen as

$$u = \sigma_2(\delta_{h2}) + \hat{d}_g + d_{g0} \tag{10}$$

where $\hat{d}_g$ is an estimate of $d_g - d_{g0}$ and updated by

$$\dot{\hat{d}}_g = \text{Proj}(\mu, \hat{d}_g) = \mu - \frac{\nu_1 \nu_2 \nabla p_d(\hat{d}_g)}{4\left(\epsilon^2 + 2\epsilon\left(d_{gm} - \epsilon\right)\right)^{n+1}\left(d_{gm} - \epsilon\right)^2} \tag{11}$$

where $\mu = k_{dg}\delta_{h2}$, $p_d(\hat{d}_g) = \hat{d}_g^2 - (d_{gm} - \epsilon)^2$, $\nabla p_d(\hat{d}_g) = 2\hat{d}_g$, n is an nonnegative integer

$$\nu_1 = \begin{cases} p_d^{n+1}(\hat{d}_g), & \text{if } p_d(\hat{d}_g) > 0 \\ 0, & \text{otherwise} \end{cases}$$

$\nu_2 = \hat{d}_g\mu + \sqrt{\left(\hat{d}_g\mu\right)^2 + \delta^2}$, and ϵ, δ are arbitrary positive constants. The function $\text{Proj}(\cdot)$ is a projection operator from [29] which is n times continuously differentiable and guarantees that $\hat{d}_g$ lies in the compact set

$$\bar{\Omega}_{dg} = \left\{\hat{d}_g : |\hat{d}_g| < d_{gm}\right\}. \tag{12}$$

Theorem III.1: The equilibrium point $[\delta_{h1}, \delta_{h2}]^T = 0$ of the system (7) with the input (10) and (11) is asymptotically stable if the control parameters satisfy

$$-(b_2k_2 - b_1k_1)\,l_2 + 2b_2d_{gm} + b_1k_1M_1 < 0 \tag{13a}$$

$$k_1l_1 > l_2 \tag{13b}$$

$$k_2l_2/2 > d_{gm} \tag{13c}$$

and the control u lies in the compact set

$$\Omega_u = \{u \in \mathbb{R} : d_{g0} - d_{gm} - M_2 \le u \le d_{g0} + d_{gm} + M_2\}.$$

Proof: Using the law (11), we have $|\tilde{d}_g| < 2d_{gm}$, where $\tilde{d}_g = d_g - \hat{d}_g - d_{g0}$. Using the control law (10), (7b) becomes

$$\dot{v}_3^a = -b_2(\sigma_2(\delta_{h2}) - \tilde{d}_g).$$

When $\delta_{h2} \ge l_2$, using (13c) we obtain $\sigma_2(\delta_{h2}) - \tilde{d}_g \ge k_2l_2 - 2d_{gm} > 0$. Hence, we have

$$v_3^a(t) \le v_3^a(0) - b_2(k_2l_2 - 2d_{gm})t. \tag{14}$$

From the definition of δ_{h2} in (8b), inequality (14) becomes

$$\delta_{h2} + \sigma_1(\delta_{h1}) \le \delta_{h2}(0) + \sigma_1(0) - b_2(k_2l_2 - d_m)t.$$

From Property 4) of σ_i, we have

$$\delta_{h2} \le \delta_{h2}(0) + 2M_1 - b_2(k_2l_2 - 2d_{gm})t.$$

Hence, when

$$t > T_2 = \frac{\delta_{h2}(0) + 2M_1 - l_2}{b_2(k_2l_2 - 2d_{gm})}$$

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δ_{h2} enters the linear region of σ_2. If the initial value of $\delta_{h2}(0) = l_2$, we have

$$\dot{\delta}_{h2} = -b_2(k_2 l_2 - \tilde{d}_g) + b_1 \frac{d\sigma_1}{d\delta_1}(\sigma_1(\delta_1) + l_2).$$

If (13a) is satisfied, we have $\dot{\delta}_{h2} < 0$. Similarly, we can show that when $\delta_{h2} < -l_2$, δ_{h2} will enter the linear region in finite time, and $\dot{\delta}_{h2} > 0$ when $\delta_{h2} = -l_2$. Hence, we conclude that δ_{h2} will enter the linear region in finite time T_{h2} and remain there for $t > T_{h2}$. Using (9a) and the fact that $|\delta_{h2}(t)| \leq l_2,\ t > T_{h2}$, it can be shown that when (13b) is satisfied, δ_{h1} will enter the linear region of σ_1 at finite time T_{h1}. Hence, after a finite time, the system (9) becomes

$$\dot{\delta}_{h1} = -b_1(k_1\delta_{h1} + \delta_{h2}) \tag{15a}$$

$$\dot{\delta}_{h2} = -b_2(k_2\delta_{h2} - \tilde{d}_g) + b_1 k_1(k_1\delta_{h1} + \delta_{h2}). \tag{15b}$$

To show the stability of the closed-loop system (15), the Lyapunov function candidate is chosen as

$$V_h = k_1^2\delta_{h1}^2/2 + \delta_{h2}^2/2 + b_2\tilde{d}_g^2/(2k_{dg}). \tag{16}$$

Using (15), the time derivative (16) is given as

$$\dot{V}_h = -b_1 k_1^3 \delta_{h1}^2 - (b_2 k_2 - b_1 k_1)\,\delta_{h2}^2 - b_2\tilde{d}_g\left(\dot{\hat{d}}_g/k_{dg} - \delta_{h2}\right).$$

From [29], the projection operator satisfies

$$\tilde{d}_g \mathrm{Proj}(\mu, \hat{d}_g) \geq \tilde{d}_g \mu.$$

Hence,

$$\dot{V}_h \leq -b_1 k_1^3 \delta_{h1}^2 - (b_2 k_2 - b_1 k_1)\,\delta_{h2}^2.$$

From inequality (13a), we have

$$b_2 k_2 - b_1 k_1 > (2b_2 d_{gm} + b_1 k_1 M_1)/l_2 > 0.$$

According to the LaSalle–Yoshizawa theorem [30, Th. 2.1] we have shown equilibrium $[\delta_{h1}, \delta_{h2}]^T = 0$ is asymptotically stable. ■

2) Lateral and Yaw Motion Control: Using (2a), (3a), and (6), the lateral subsystem using point image moment features is

$$\dot{s}_{lp} = -\lambda b_1 v_l^a - S s_{lp}\dot{\psi} \tag{17a}$$

$$\dot{v}_l^a = b_2\bar{u} - S v_l^a \dot{\psi} \tag{17b}$$

where $\bar{u} = uS\eta_1$ is the input and $\eta_1 = [\phi, \theta]^T$. For the lateral subsystem (17), the control law is chosen as

$$\bar{u} = -\Sigma_3\left(v_l^a - k_4 h(\beta s_{lp})\right) \tag{18}$$

where k_4, β are positive constants and h is defined in (5). The derivative of $h(\beta s_{lp})$ at zero is proportional to β. The parameter β is used to tune the convergence rate of the closed-loop. The yaw control for (2c) is

$$\dot{\psi} = K_\psi s_{\psi p}. \tag{19}$$

Hence, the closed-loop for yaw becomes

$$\dot{s}_{\psi p} = -K_\psi s_{\psi p}. \tag{20}$$

Theorem III.2: The equilibrium $[s_{lp}, v_l^a, s_{\psi p}]^T = 0$ of system (17), (2c) with input (18) and (19) is asymptotically stable if the controller parameters satisfy

$$k_4 < l_3/2 \tag{21}$$

$K_\psi, k_3 > 0$, and the magnitude of each element of the control input is less than M_3.

Proof: The first step is to show that the saturation function Σ_3 will enter its linear region in finite time. To prove this, we introduce an energy storage function

$$V_{l2} = (v_l^a)^T v_l^a/2.$$

Using the control law (18) and (17b), the time derivative of V_{l2} is

$$\dot{V}_{l2} = -b_2 v_l^a \Sigma_3\left(v_l^a - k_4 h(\beta s_{lp})\right).$$

From the Lemma III.1, we can conclude that $\dot{V}_{l2} < 0$ when $\|v_l^a\| > l_2/2$ and (21) is satisfied. Hence, the saturation function Σ_3 enters its linear region in finite time. The closed-loop system becomes

$$\dot{s}_{lp} = -\lambda b_1 v_l^a - S s_{lp}\dot{\psi}$$

$$\dot{v}_l^a = b_2\left(-k_3 v_l^a + k_3 k_4 h(\beta s_{lp})\right) - S v_l^a \dot{\psi}.$$

The Lyapunov function candidate is chosen as

$$V_l = \frac{1}{\lambda b_1 \beta}\left(\sqrt{1 + \beta^2 s_{lp}^T s_{lp}} - 1\right) + \frac{1}{2b_2 k_3 k_4}(v_l^a)^T v_l^a$$

whose time derivative is

$$\dot{V}_l = -(v_l^a)^T v_l^a/k_4. \tag{22}$$

To control yaw motion, we consider the Lyapunov function candidate

$$V_s = V_l + s_{\psi p}^2/2$$

Using (20) and (22), the time derivative of V_s is

$$\dot{V}_s = -(v_l^a)^T v_l^a/k_4 - K_\psi s_{\psi p}^2.$$

Hence, using LaSalle's invariance theorem we conclude that the equilibrium $[s_{lp}, v_l^a, s_{\psi p}]^T = 0$ of the closed-loop is asymptotically stable. ■

IV. Simulation and Experimental Results

A. Experimental Platform

We validate the proposed control on the indoor Applied Nonlinear Control Lab (ANCL) quadrotor platform. The ANCL quadrotor UAV is shown in Fig. 5. The ANCL quadrotor UAV consists of a 3-D robotics vehicle frame, a CMUcam5 Pixy onboard computer vision system [31], and a PX4 autopilot [32]. The PX4 contains a 3-D accelerometer, a 3-D gyroscope, and a 3-D magnetometer. An extended Kalman filter-based AHRS fuses the PX4 sensor data to provide an estimate of attitude and angular velocity. The PX4 computes the proposed IBVS law and generates the PWM signals for the four ESCs that power four 1100 kV brushless motors with 11 × 4.5 inch multirotor

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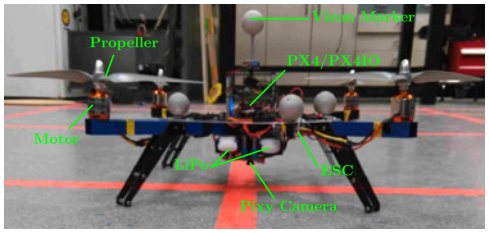

Fig. 5. ANCL quadrotor UAV.

TABLE I
QUADROTOR AND CAMERA MODEL PARAMETERS

m	1.6 kg
J	$\mathrm{diag}([0.03, 0.03, 0.05])$ kg· m^2
λ	2.8 mm
Image sensor size	1/4″
Image resolution	320 × 200 pixels
FoV	75° horizontal and 47° vertical
ℓ	0.25 m

propellers. The Pixy camera is mounted on the bottom of the quadrotor and connected to the PX4 via a serial connection. It transmits image point features at a rate of about 25 Hz with a resolution of 320 × 200 pixels. The lens mounted on the Pixy has 75° horizontal and 47° vertical FoV. An eight-camera Bonita Vicon motion capture system measures and sends the vehicle's translational velocity to the PX4 via a radio at about 100 Hz. The Vicon system has millimetre position accuracy. Power is supplied by two 11.1 V 2600 mAh LiPo batteries. The entire vehicle system weight including the battery and the computer vision system is 1.6 kg. Table I summarizes the quadrotor and camera model parameters.

B. Simulation Results

We use the model parameters in Table I for simulation and take $K_T = 40$ N/ms^2. The desired value of the image feature $s = [s_{lp}, s_{hp}, s_{\psi p}]^T$ is chosen as $s^* = [0, 0, 1, 0]$, and the image feature error denoted as $e_s = [e_{s1}, e_{s2}, e_{s3}, e_{s4}]^T = s - s^*$. The desired 3-D translational displacement in $\mathcal{N}$ is $d_{nc}^{n*} = [0,0,-1.5]^T$ m, which corresponds to the value of d_{nc}^n when $e_s = 0$. The initial displacement of the vehicle in $\mathcal{N}$ is $[0.55,-0.55,-2.3]^T$ m with zero roll and pitch, and the initial value of yaw is $\pi/3$ rad. The initial velocities are $v^a(0) = [-0.4,-0.4,0.2]^T$ m/s and $\omega^c(0) = 0$. The visual target consists of two points located at coordinates $[-0.18,0,0]^T$ m and $[0.18,0,0]^T$ m in $\mathcal{N}$. The saturation function σ_i is chosen as

$$\sigma_i(x) = \begin{cases} k_i x, & \text{if } |x| < l_i \\ \mathrm{sign}(x)\left(k_i l_i + c_{1i} \arctan\left(c_{2i}\left(|x| - l_i\right)\right)\right), & \text{otherwise} \end{cases} \tag{23}$$

where $c_{1i} = 2(M_i - k_i l_i)/\pi$ and $c_{2i} = k_i/c_{1i}$. Fig. 6 shows the shape of σ_i. The parameters for the saturation functions σ_i, $i = 1, 2, 3$, are given in Table II. The other control parameters are selected as $k_4 = 0.06, k_{dg} = 0.04, K_\psi = 0.1$, and the

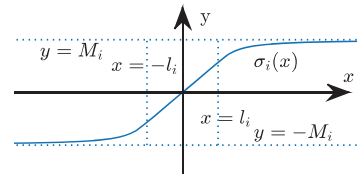

Fig. 6. Saturation function $\sigma(x)$.

TABLE II
PARAMETERS FOR SATURATION FUNCTIONS σ_i, $i = 1, 2, 3$

Gains	k_i	l_i	M_i
$i = 1$	1	0.5	0.55
$i = 2$	0.4	0.42	0.17
$i = 3$	0.5	0.16	0.1

parameters in (11) are chosen as $n = 1, \epsilon = 0.001$, and $\delta = 0.01$. From the value of d_{nc}^{n*}, we get $X_3^{a*} = 1.5$ m. From the value of K_T and m, we obtain $d_g = 0.3924$. The initial value of $\hat{d}_g$ is 0.38, i.e., $d_{g0} = 0.38$, and the estimate $\hat{d}_g$ is limited to set Ω_{dg} defined in (12) with $d_{gm} = 0.04$. It is straightforward to show that these control gains satisfy (13) and (21).

To verify that the input saturation helps keep the visual target in the FoV of the camera, we considered two cases in the simulation. In the first case, we applied the input saturation control (10) and (18). In the second case, we removed all saturation functions in (10) and (18). To do this, we choose the linear region expression of (10)

$$u = k_2 \delta_{h2l} + d_{g0} + \hat{d}_{gl} \tag{24}$$

where $\delta_{h2l} = v_3^a - k_1 \delta_{h1}$, and $\hat{d}_{gl}$ is updated by

$$\dot{\hat{d}}_{gl} = k_{dg} \delta_{h2l}. \tag{25}$$

It can be easily verified that the origin of the closed-loop system (7) with inputs (24) and (25) are globally exponentially stable. To remove the saturation function from controller (18), we linearize (18) at the origin and obtain

$$\bar{u} = -k_3(v_l^a - k_4 \beta s_{\mathrm{lp}}). \tag{26}$$

We readily prove that the origin of system (17) with input (26) is exponentially stable.

Fig. 7 shows the trajectories of e_s, $e_d = d_{nc}^n - d_{nc}^{n*}$, and $e_\psi = \psi - \psi^*$ for both controllers, where ψ^* is the desired yaw of the vehicle. In both cases, the image feature error and the position error converge to zero. The difference between the two cases is that error signals e_s and e_d asymptotically converge with controller (10) and (18), whereas exponential convergence is obtained using controllers (24) and (26). As expected, the control signal u lies in $\Omega_u = \{u : 0.17 \le u \le 0.59\}$ using the control (10) and (18). Fig. 8(a) shows the trajectory of the control signal u, and it remains in Ω_u as expected. The dotted line in Fig. 8(a) denotes the boundary of Ω_u. Fig. 8(b) shows the trajectory of u with control (24) and (26). In this case, u leaves Ω_u.

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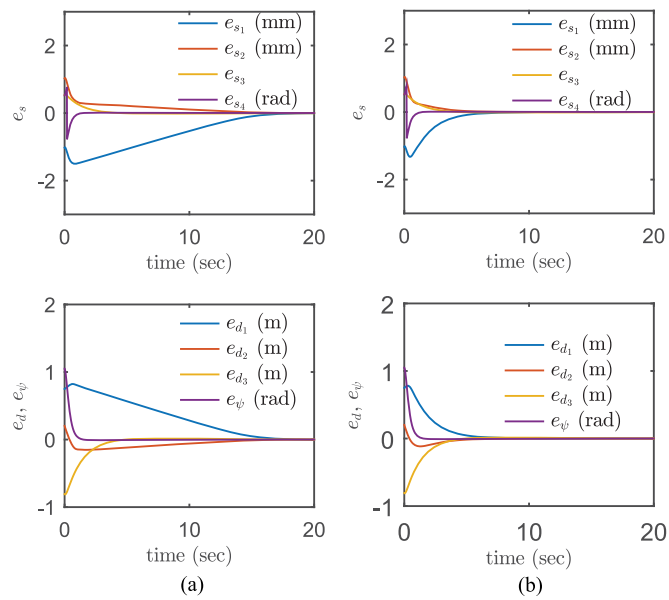

Fig. 7. Trajectories of image feature error e_s, vehicle's 3-D position error e_d, and e_ψ. (a) With input saturation and (b) without input saturation.

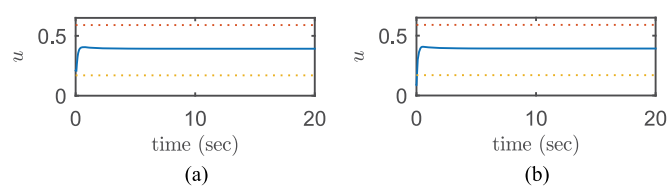

Fig. 8. Trajectories of the input signal u. (a) With input saturation and (b) without input saturation.

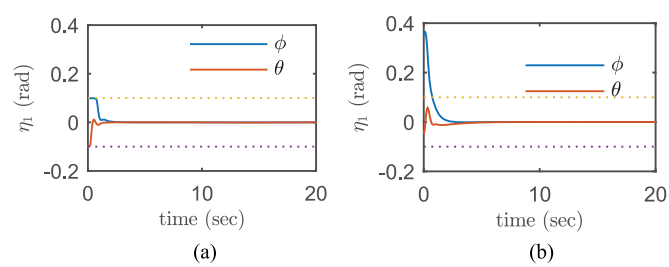

Fig. 9. Trajectories of the input signals ϕ and θ. (a) With input saturation and (b) without input saturation.

Fig. 9 shows the trajectories of the reference input η_1 for the outer loop for the two cases. It can be seen that the input signal, i.e., the roll and pitch lie in $\Omega_\phi = \{\phi : |\phi| \leq 0.1\}$ and $\Omega_\theta = \{\theta : |\theta| \leq 0.1\}$ when input saturation is considered. The reference roll and pitch signal leave Ω_ϕ and Ω_θ using (24) and (26). Fig. 10 shows the image for the two cases where the rectangle denotes the boundary of the camera's image plane. It can be seen that when input saturation is considered the image is kept in the FoV of the camera. The image of one of the two points leaves the FoV of the camera when input saturation is off.

C. Experimental Results

The experiment considers a target consisting of two points: $[0.2,0,0]^T$ m and $[-0.2,0,0]^T$ m. The saturation function σ_i is

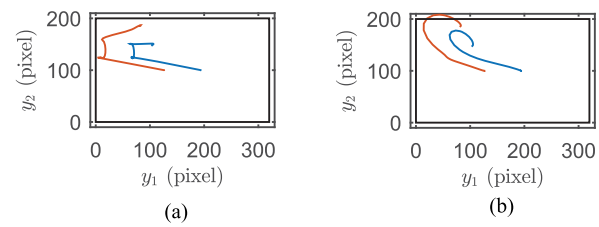

Fig. 10. Image trajectories of the target points. (a) With input saturation and (b) without input saturation.

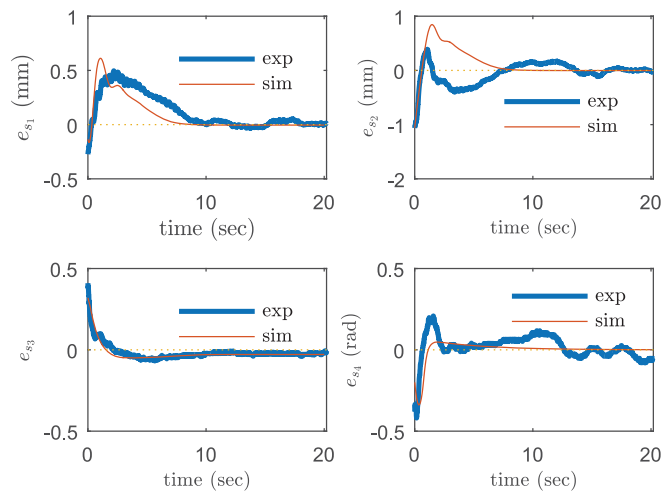

Fig. 11. Experimental trajectories of image feature error e_s.

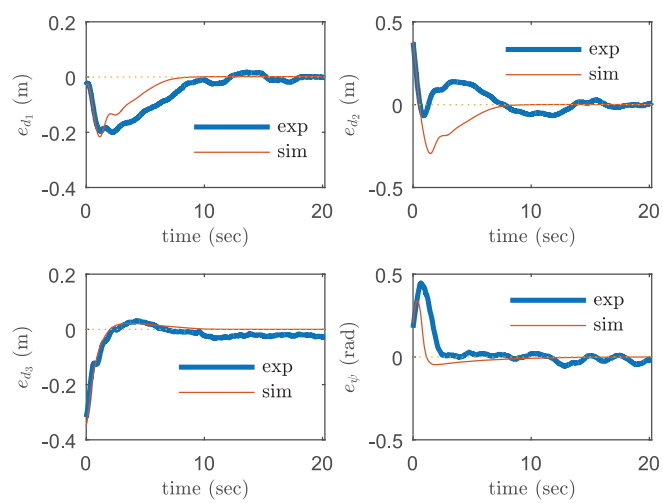

Fig. 12. Experimental trajectories of vehicle's 3-D position error e_d and yaw error e_ψ.

(23) which is the same as the one used in simulation. The values of the control parameters k_i, l_i, M_i, $i = 1, 2, 3$, k_{dg}, k_4, K_ψ, ϵ, δ, and d_{gm} are the same, as mentioned in Section IV-B. The inner loop attitude control is the PID controller given in [15]. Fig. 11 shows the trajectory of image feature error signal e_s. The corresponding 3-D trajectories of translational error $e_d = [e_{d_1}, e_{d_2}, e_{d_3}]^T = d_{nc}^n - d_{nc}^{n*}$ and yaw error $e_\psi = \psi - \psi^*$ are shown in Fig. 12. Simulation results with appropriately chosen initial conditions were superimposed on the experimen-

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TABLE III
STATISTICS OF e_s, e_d, AND e_ψ

Image Feature	Mean	Standard Deviation
e_{s_1} (mm)	0.04	0.04
e_{s_2} (mm)	0.01	0.05
e_{s_3}	– 0.02	0.01
e_{s_4} (rad)	0.01	0.03
e_{d_1} m	– 0.014	0.016
e_{d_2} (m)	– 0.006	0.015
e_{d_3} (m)	0.028	0.011
e_ψ (rad)	0.004	0.019

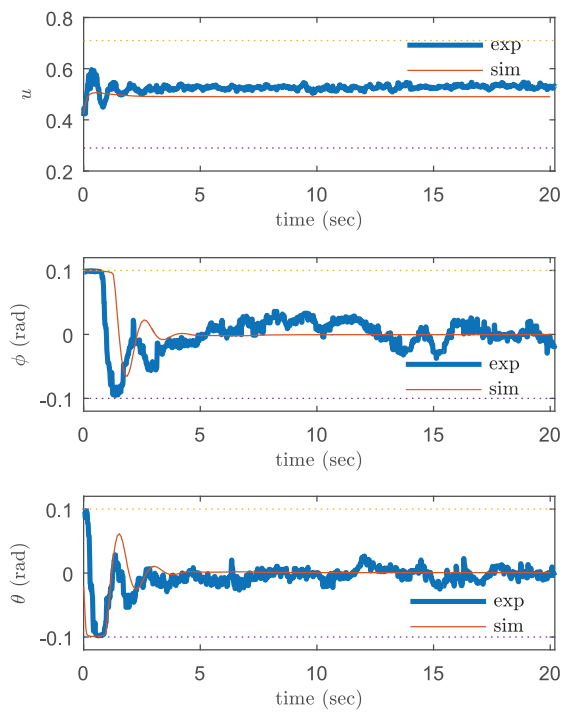

Fig. 13. Experimental trajectories of outer loop inputs.

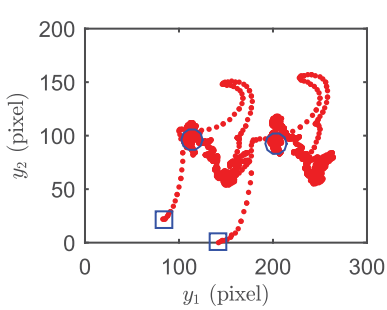

Fig. 14. Image points trajectories.

tal results in Figs. 11 and 12. We observe that the results are close. This illustrates that the model of the closed-loop system analyzed in Section III is accurate, and justifies the approximations made in the open-loop modeling in Section II-A. We assume

$$d_{nc}^{n*} = [0.0766, -0.0781, -1.1399]^T\, m$$

and $\psi^* = 1.1567$ rad. In order to measure steady-state performance of the controller, we compute the time at which the error signals e_s and e_d reach steady-state. We denote this steady-state time by t_{ss}. The value of t_{ss} is obtained by computing the difference in mean values of the error signals in the two intervals $[t_{ss}, t_e]$ and $[(t_{ss}+t_e)/2, t_e]$, where $t_e = 40$ s is the time at which the IBVS law is turned off. The value of t_{ss} is determined by the smallest time after which this difference remains negligible. Using this method we obtain $t_{ss} = 14$ s. The average and standard deviation of e_s, e_d, and e_ψ in steady-state, i.e., on the time interval $[t_{ss}, t_e] = [14{,}40]$ s are given in Table III. The small values demonstrate steady-state tracking performance of the proposed control. Although small, we remark that steady-state errors are still nonzero due to unknown attitude measurement bias. One approach to account for this bias is given in [15], but there input constraints are not necessarily satisfied. Steady-state error of the proposed control is significantly smaller than that of the PD control given in [15], which does not compensate for bias. This is because the proposed saturation controller provides high gain when the error signal is close to zero, and this reduces steady-state error.

The experimental and simulated trajectories of the control input for the outer loop are shown in Fig. 13. As expected, u, the reference roll and pitch signals lie in Ω_u, Ω_ϕ, and Ω_θ. Fig. 14 shows the image points trajectories, where the square and circle markers denote the initial and final image points, respectively. As can be seen, the trajectories remain in the camera FoV.

V. CONCLUSION

An input saturation visual servoing law is proposed to control the height, lateral, and yaw motion of a quadrotor UAV. The control law is proven to asymptotically stabilize the closed-loop. The control law for the height subsystem restricts thrust to a predefined compact set. The positive thrust is guaranteed by setting a positive lower bound for the compact set. The control law for the lateral subsystem bounds the reference roll and pitch to a compact set. This avoids large roll and pitch when the image feature error is large, and this helps to keep the visual target in the camera FoV. The saturation controller proposed in this paper provides high gain when the image feature error approaches zero and therefore reduces the steady-state error of the image feature. Both simulation and experimental work validate the performance of the proposed input saturation control on the ANCL quadrotor test stand. Future work includes generalizing the method to moving and nonhorizontal targets. We plan to develop the theory to explicitly account for attitude bias input disturbances to the lateral subsystem. Improvements to the onboard vision computer are planned which should improve the performance of the outer loop. For example, reduced image processing delay leads to improved outer loop bandwidth.

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